

Chapter One

Energy Reserves and Potential

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CHAPTER 1: Energy Reserves and Potential

Introduction

Energy reserves are part of the energy resources that, based on technical, economic and other relevant (e.g. environmental) considerations, can be recovered and for which extraction is justified.

The exact definition of reserves depends on the kind resources in focus.

Global Classification of Energy Reserves

The classification of energy reserves is guided by the United Nations Framework Classification for Fossil Energy and Mineral Reserves and Resources (UNFC 2009).

This framework evaluates resources based on: Economic and Social Viability, Field Project Status and Feasibility, and Geological Knowledge.

- Class A: Commercially recoverable resources.
- Class B: Potentially commercially recoverable resources.
- Class C: Non-commercial and other known deposits.

Energy Reserves in India

India has a diverse range of energy resources, with significant reserves of both fossil fuels and renewable energy potential.

The country's energy mix is undergoing a transformation to reduce dependence on fossil fuels and promote the use of renewable energy sources, driven by environmental, economic, and macroeconomic consideration.

This chapter highlights the energy reserves and potential in India, focusing on both fossil fuels and renewable energy resources. It explores coal, lignite, crude oil, natural gas, and renewable energy potential, providing an overview of the country's energy landscape.

Estimated Reserves of Coal in India as on 01.04.2025

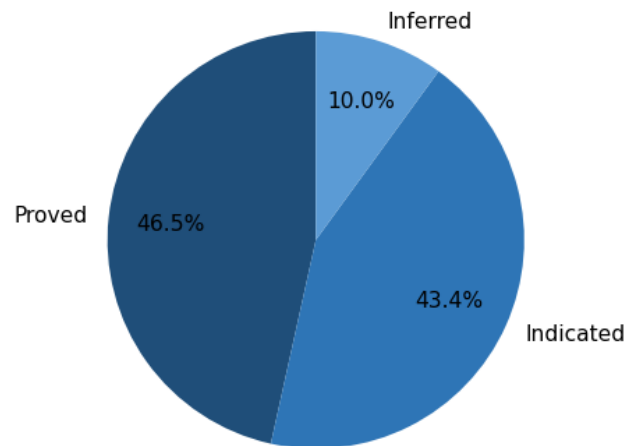


Fig 1.1 Estimated Reserves of Coal in India as on 01.04.2025

Estimated Reserves of Lignite in India as on 01.04.2025

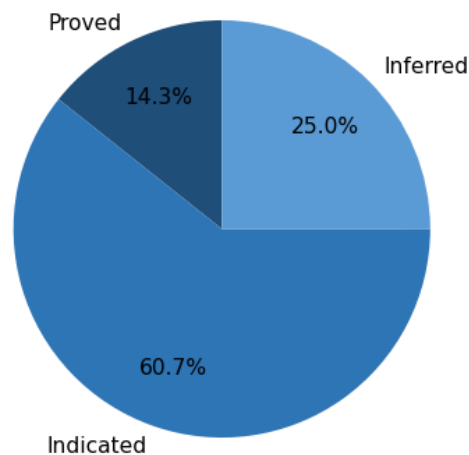


Fig 1.2 Estimated Reserves of Lignite in India as on 01.04.2025

1.1 Coal Reserves

India is home to one of the largest proven coal reserves in the world. As of 01-04-2025, the total estimated coal reserves were 400.7 billion tonnes, showing an addition of 11.29 billion tonnes compared to the previous year.

This marks a growth of 2.90% in total coal reserves during the year 2024-25(P) over the previous year (2023-24) (Table 1.1).

The distribution of coal reserves in India is concentrated in a few states like Odisha, Jharkhand and Chhattisgarh. These three states account for approximately 70% of the total coal reserves in the country. Out of the total reserves in the country, proven reserves i.e. economically viable and geologically confirmed account for approximately 55% of the total estimated coal reserves as depicted in the figure 1.1.

1.2 Lignite Reserves

1.3 Crude Oil Reserves

As of 01-04-2025, the estimated reserves of crude oil in India stood at 672.07 million tonnes, compared to 671.40 million tonnes in the previous year, reflecting a 0.10% increase over the last year (Table 1.3).

Geographically, the maximum crude oil reserves in India are concentrated in the Western Offshore region, which accounts for 31% of the total crude oil reserves. This is followed by the Assam region, which holds around 22% of the country's crude oil reserves as shown in figure 1.3.

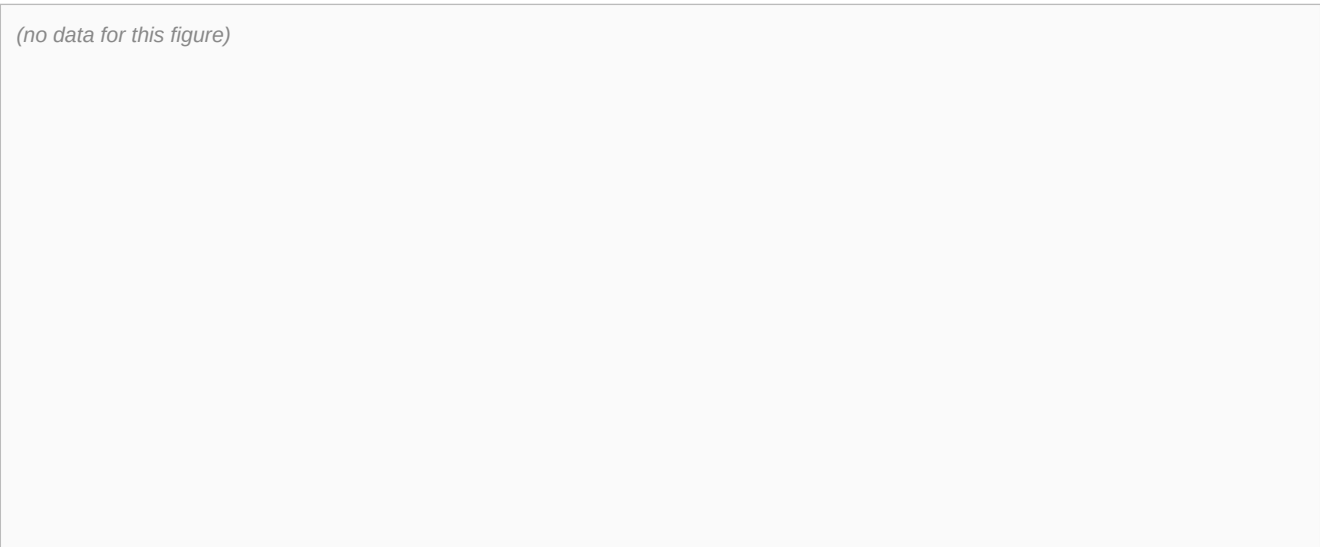


Fig 1.3: Estimated Reserves of Crude Oil in India as on 01.04.2025

1.4 Natural Gas Reserves

The estimated reserves of natural gas as of 01-04-2025 were 1,073.01 Billion Cubic Meters (BCM) compared to 1,094.19 Billion Cubic Meters (BCM) in the previous year, reflecting a decrease of 1.94% over the last year (Table 1.4).

The largest reserves of natural gas in India are located in the Western Offshore region, which holds approximately 32% of the total natural gas reserves. This is followed by the Eastern Offshore, which accounts for 25% of the reserves as shown in figure 1.4.

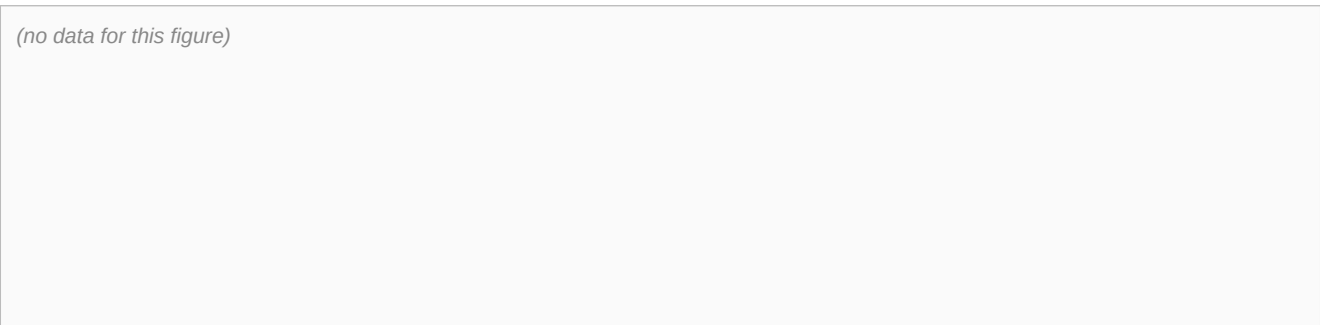


Fig 1.4: Estimated Reserves of Natural Gas in India as on 01.04.2025

1.5 Renewable Energy Potential in India

India is increasingly focusing on renewable energy sources as part of its sustainable development and mitigating impact on climate. As of March 31, 2025, the total estimated potential for renewable power generation in India stood at 47,04,043 MW.

This potential comes from a variety of renewable sources, including wind, solar, biomass, small hydro, and cogeneration from bagasse (Table 1.4). Figure 1.5 shows the potential is spread across several key renewable energy sources, as detailed below:

Source wise Estimated Potential of Renewable Power in India as on 31.03.2025

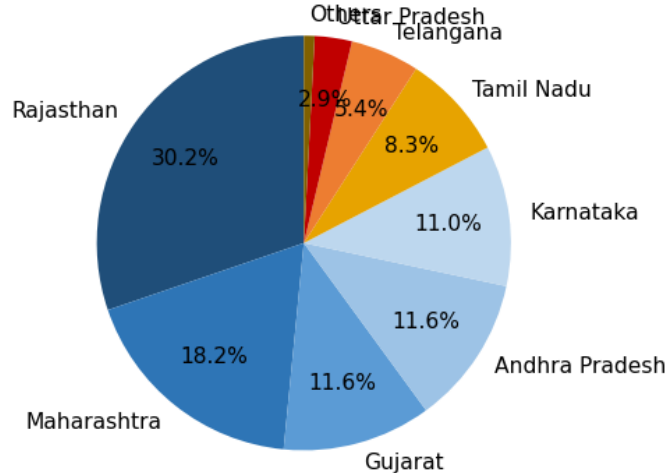


Fig 1.5: Source wise Estimated Potential of Renewable Power in India as on 31.03.2025

Solar Power: India has immense potential for solar power generation, estimated at 33,43,378 MW, which accounts for 71% of the total renewable energy potential. Solar power stands as the most significant source of renewable energy in India.

Wind Power: With an estimated potential of 1,163,856 MW (close to 25% of total renewable energy potential), it has the potential to contribute significantly in shaping the renewable energy-generation scenario of India.

Large Hydropower (Large Hydro): The potential for large hydropower generation is estimated at 133,410 MW, contributing 2.84% to the country's renewable energy potential.

Small Hydro Power (SHP): India also has a notable potential of 21,134 MW (0.45%) from small hydro power projects.

Biomass Power: Biomass, which includes agricultural waste, forest residues and other organic matter, has a potential of 28,447 MW, accounting for 0.60% of the total renewable power potential.

Cogeneration from Bagasse: India has a specific potential of 13,818 MW (0.29%) from bagasse-based

Geographical Distribution of Renewable Energy Potential

The geographical distribution of renewable energy potential in India (Figure 1.6) shows significant regional variations. The states with the highest renewable energy potential, as of March 31, 2025, are: Rajasthan, Maharashtra, Gujarat, Andhra Pradesh, and Karnataka.

Together, these five states account for around 65% of India's total renewable energy potential, highlighting their critical role in India's renewable energy future.

Statewise Estimated Potential of Renewable Power in India as on 31.03.2025

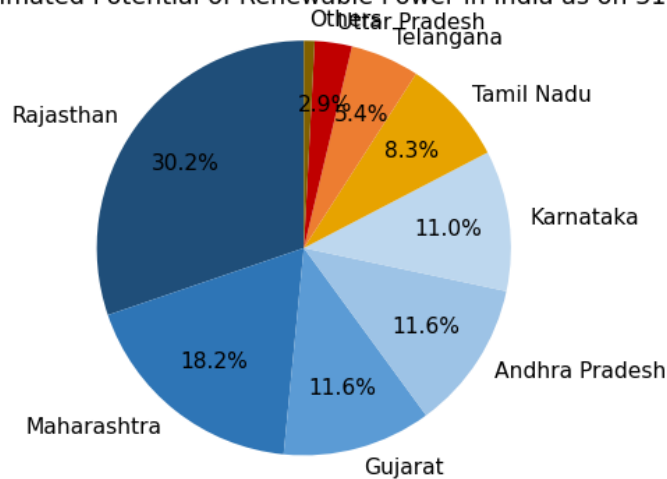


Fig 1.6: Statewise Estimated Potential of Renewable Power in India as on 31.03.2025

Table 1.1: Statewise Estimated Reserves of Coal (As on 1st April 2025)

States/UTs	Proved...	Proved...	Indicat...	Indicat...	Inferre...	Inferre...	Total 2...	Total 2...	Distrib...	Distrib...
Jharkhand	45,563	45,563	31,439	31,439	6,559	6,559	83,561	83,561	—	—
Odisha	37,391	37,391	34,165	34,165	9,410	9,410	80,966	80,966	—	—
Chhattisgarh	20,460	20,460	34,433	34,433	2,228	2,228	57,121	57,121	—	—
West Bengal	13,403	13,403	13,022	13,022	4,880	4,880	31,305	31,305	—	—
Madhya Pradesh	11,958	11,958	12,176	12,176	3,848	3,848	27,982	27,982	—	—
Telangana	10,486	10,486	8,511	8,511	2,683	2,683	21,680	21,680	—	—
Maharashtra	7,286	7,286	3,060	3,060	1,956	1,956	12,302	12,302	—	—
Total	146,547	146,547	136,806	136,806	31,564	31,564	314,917	314,917	—	—

Table 1.2: Statewise Estimated Reserves of Lignite(As on 1st April 2025)

States/UTs	Proved...	Proved...	Indicate...	Indicate...	Inferred...	Inferred...	Total 2024	Total 2025	Distribut...
Tamil Nadu	4,262	4,262	24,683	24,683	8,376	8,376	37,321	37,321	—
Gujarat	1,278	1,278	283	283	1,161	1,161	2,722	2,722	—
Rajasthan	1,168	1,168	3,029	3,029	2,138	2,138	6,335	6,335	—
Jammu & Kashmir	0	0	20	20	7	7	27	27	—
Kerala	0	0	0	0	10	10	10	10	—
Puducherry	0	0	405	405	12	12	417	417	—
Total	6,708	6,708	28,420	28,420	11,704	11,704	46,832	46,832	—

Table 1.3: Statewise Estimated Reserves of Crude Oil and Natural Gas (As

States/UTs/Region	Crude Oil 2024	Crude Oil 2025	Natural Gas 2024	Natural Gas 2025
Andhra Pradesh	7.69	8.41	59.27	60.60
Arunachal Pradesh	2.88	0.44	7.33	7.10
Assam	145.41	144.73	164.51	163.33
Gujarat	118.86	121.20	55.46	54.74
Nagaland	2.38	2.38	0.09	0.09
Rajasthan	131.50	136.46	63.55	27.12
Tamil Nadu	8.54	9.11	36.57	38.39
Tripura	0.07	0.07	28.18	27.73
West Bengal	0.15	0.20	0.00	0.79
West Bengal (CB...	0.00	0.00	44.37	44.93
Jharkhand (CBM)	0.00	0.00	4.86	5.67
Madhya Pradesh...	0.00	0.00	24.38	23.15
Eastern Offshore	40.67	37.25	263.34	270.67
Western Offshore	213.26	209.30	342.29	348.70
Total	671.40	672.07	1094.19	1073.01

Table 1.4: Sourcewise and Statewise Estimated Potential of Renewable

States/UTs	Wind Pow...	Small Hyd...	Biomass...	Cogenera...	Solar Ene...	Large Hydro	Total	Distributi...
Andhra Pradesh	409	123336	1999	280	299312	2596	427933	9.10
Arunachal Pradesh	246	18	2065	0	468	50394	53191	1.13
Assam	202	459	322	0	19173	643	20799	0.44
Bihar	964	527	4023	347	32991	130	38983	0.83
Chhattisgarh	1098	2749	354	0	126484	1311	131996	2.81
Goa	33	14	5	0	6752	0	6804	0.14
Gujarat	180790	202	2638	555	243220	550	427954	9.10
Rajasthan	284250	1300	52	0	828781	411	1114794	23.70
Maharashtra	173868	786	2630	3917	486679	3144	671024	14.26
All India Total	1163856	21134	28447	13818	3343378	133410	4704043	100.00